

HEALIX

AI-Powered Personalized Health & Lifestyle Assistant

Updated MVP Document — Post Phase 1

1. Project Vision

People with chronic lifestyle conditions like Polycystic Ovary Syndrome (PCOS), Type 2 Diabetes, Hypothyroidism, Obesity, and Hypertension need personalised lifestyle management. Currently they must juggle one app for calorie tracking, another for workouts, ChatGPT for meal plans, Google for recipes, and a notes app for progress — everything is fragmented.

Healix combines all of the following inside a single platform:

- Health profile management
- AI-generated meal plans
- AI-generated workout plans
- Progress tracking
- Condition-specific recommendations
- Habit building

2. MVP Scope

For the first version, only PCOS is supported because:

- You understand the problem domain personally
- Huge target audience
- Sufficient medical complexity to validate the AI layer
- Easier dataset collection compared to other conditions

Do NOT add Diabetes, Thyroid, or Hypertension initially.

3. Updated User Flow (Post Phase 1)

The original MVP described a linear five-step flow. After Phase 1 development, this has been expanded into a richer onboarding experience with a multi-step wizard, additional data collection, and immediate health insights — all before any AI layer exists.

3.1 Original Flow vs. Updated Flow

Original Flow	Updated Flow (Phase 1 Delivered)
1. User signs up	1. Landing Page
2. Profile Setup (single form)	2. Signup Page (with full validation)
3. Condition Selection	3. Login Page
4. AI Generates Plans	4. Onboarding Step 1: Basic Information
5. Dashboard	+ Step 2: Nutrition Profile (NEW)
	+ Step 3: Fitness & Lifestyle (NEW)
	+ Health Summary Page (NEW)
	+ Dashboard with live health cards (EXPANDED)

4. Detailed Phase 1 Changes

Change 1 — Authentication Pages

The original document listed Login and Signup as the only auth pages. Phase 1 delivered a full three-page auth system with client-side validation.

Original MVP	Updated After Phase 1
Login page	+ Landing Page
Signup page	Login Page
	Signup Page
	+ Form validation
	+ Email format validation
	+ Password length validation
	+ Password match validation

+ Navigation between pages

Change 2 — Profile Setup to Multi-Step Wizard

The original document described a single profile form. This was replaced with a four-step onboarding wizard to improve UX, reduce cognitive load, and increase completion rates.

Original: One Giant Form	Updated: 4-Step Wizard
Age, Gender, Height, Weight, Goal Weight, Activity Level, Food Preference, Budget, Lifestyle — all on one screen	Step 1 — Basic Information Step 2 — Nutrition Profile Step 3 — Fitness & Lifestyle Step 4 — Health Summary

Rationale:

- Better UX — users are never overwhelmed
- Higher completion rate — progressive disclosure
- More professional product feel
- Easier to add or remove fields per step in future

Change 3 — Expanded User Data Collection

Five new data fields were added that were not in the original document.

Original MVP	Updated After Phase 1
Age	Age
Gender	Gender
Height	Height
Weight	Weight
Goal Weight	Goal Weight
Activity Level	Food Preference
Food Preference	+ Favourite Foods
Budget	+ Disliked Foods
Lifestyle	+ Dietary Restrictions
	Monthly Food Budget
	Activity Level

	+ Available Equipment + Occupation
	Lifestyle Description

Change 4 — Immediate Health Insights (Pre-AI)

Three health metrics are now calculated and displayed at the end of onboarding without any AI call, using standard medical formulas applied to the user's input.

Metric	Formula Used	Where Shown
BMI	weight (kg) / height squared (m)	Step 4 Summary + Dashboard
Daily Water Intake	weight x 0.033 litres	Step 4 Summary + Dashboard
Estimated Daily Calories	Mifflin-St Jeor x activity multiplier	Step 4 Summary + Dashboard

Change 5 — Dashboard Expansion

The original document described a generic dashboard with placeholder cards. Phase 1 delivered six live, data-driven cards populated from the onboarding wizard.

Original MVP	Updated After Phase 1
Today's calories (placeholder)	+ BMI Card (calculated)
Water intake (placeholder)	+ Daily Calories Card (calculated)
Workout (placeholder)	+ Water Intake Card (calculated)
Weight trend (placeholder)	+ Food Preference Card (from profile)
	+ Activity Level Card (from profile)
	+ Goal Weight Card (from profile)

Change 6 — Data Persistence via localStorage

The original Phase 1 was scoped as static UI only, with no data persistence. Phase 1 implemented client-side persistence using the browser's localStorage API.

Original Phase 1 Scope	What Was Built
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Static pages — no data stored	localStorage stores full user profile as JSON
Data lost on navigation	Data survives page refresh and navigation
Database planned for Phase 4	JSON.stringify() and JSON.parse() used for serialisation

This advancement means Phase 4 (MongoDB) can now focus on server-side sync and user accounts rather than basic persistence.

Change 7 — Condition Selection Merged into Onboarding

The original document had Condition Selection as a separate Step 3 in the user flow. Since the MVP supports only PCOS, a dedicated screen adds friction without value. In the updated flow, PCOS is pre-selected and embedded inside the onboarding wizard, removing one unnecessary step.

5. New Features Invented During Phase 1

The following features were not mentioned anywhere in the original document. They were designed and built during Phase 1:

- ✓ Multi-Step Onboarding Wizard
- ✓ Progressive Form Disclosure (4 steps)
- ✓ Favourite Foods field
- ✓ Disliked Foods field
- ✓ Dietary Restrictions field
- ✓ Available Gym Equipment field
- ✓ Occupation field
- ✓ BMI auto-calculation
- ✓ Daily Water Intake auto-calculation
- ✓ Estimated Daily Calories auto-calculation (Mifflin-St Jeor)
- ✓ Landing Page with product messaging
- ✓ Client-side data persistence via localStorage

6. Intentionally Deferred (Future Phases)

The following items from the original roadmap were not built in Phase 1 and remain on schedule:

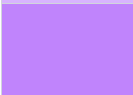
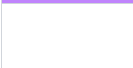
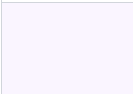
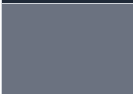
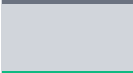
- ☐ React (Phase 2)
- ☐ FastAPI backend (Phase 3)

- ❑ MongoDB database (Phase 4)
- ❑ JWT Authentication (Phase 5)
- ❑ AI Meal Plan generation — Gemini/OpenAI (Phase 6)
- ❑ AI Workout Plan generation (Phase 6)
- ❑ Progress tracking and weight logs (Phase 7)
- ❑ Charts and analytics (Phase 7)
- ❑ Deployment — Vercel / Render / MongoDB Atlas (Phase 8)

7. Healix Design System (Phase 1)

All pages built in Phase 1 follow this design system. It defines every colour, typography rule, spacing value, and component style used across the product. Future phases must stay consistent with these tokens.

7.1 Color Palette

Swatch	Hex	Name	Purpose
	#D8B4FE	Primary Lavender	Primary buttons, logo accents, active navigation items, important highlights
	#C084FC	Hover Lavender	Button hover states, interactive cards, link hover states
	#FFFFFF	White	Cards, forms, main content sections
	#FAF5FF	Light Lavender	Page background, section backgrounds, soft contrast areas
	#1F2937	Dark Gray	Headings, paragraphs, form labels (primary text)
	#6B7280	Mid Gray	Subtitles, helper text, placeholders (secondary text)
	#D1D5DB	Border Gray	Input field borders, card borders, dividers
	#10B981	Success Green	Success messages, completed steps, positive feedback
	#EF4444	Error Red	Validation errors, form warnings, critical alerts

7.2 Typography

Element	Font Weight	Usage
Headings	700 — Bold	Page titles, section titles
Body Text	400 — Regular	Paragraphs, descriptions, labels
Buttons	600 — Semibold	All button text

7.3 Component Tokens

Token	Value	Applied To
Border Radius — Cards	12px	All card components
Border Radius — Buttons	8px	All button elements
Border Radius — Inputs	8px	All form input fields
Card Shadow	0 4px 10px rgba(0,0,0,0.1)	All card components
Border Color	#D1D5DB	Inputs, cards, dividers

7.4 Theme Support (Future)

The design system is built to support both Light and Dark themes. Brand Lavender (#D8B4FE) remains consistent across all future versions to maintain brand recognition.

Light Theme (Current)	Dark Theme (Future)
Background: #FFFFFF Text: #1F2937 Primary: #D8B4FE	Background: #121212 Text: #FFFFFF Primary: #D8B4FE

8. Phase 1 Final Output

8.1 Pages Implemented

#	Page	Key Elements
1	Landing Page	Hero section, CTA buttons, product messaging
2	Login Page	Email and password fields, form validation, link to signup

#	Page	Key Elements
3	Signup Page	Name, email, password fields, email and password validation
4	Profile Step 1 — Basic Info	Age, gender, height, weight, goal weight
5	Profile Step 2 — Nutrition	Food preference, favourite foods, disliked foods, dietary restrictions, budget
6	Profile Step 3 — Fitness	Activity level, available equipment, occupation, lifestyle description
7	Health Summary Page	Calculated BMI, water intake, daily calories before dashboard
8	Dashboard	6 live data-driven cards populated from onboarding profile

8.2 Features Implemented

Feature	Status
Form validation	Complete
Email validation	Complete
Password confirmation	Complete
Multi-step onboarding	Complete
localStorage data persistence	Complete
BMI calculation	Complete
Water intake calculation	Complete
Daily calorie estimation	Complete
Dashboard cards	Complete
Navigation flow	Complete
Responsive layout	Complete
GitHub integration	Complete

8.3 Additional Improvements Beyond Original MVP

Improvement	Category
Favourite foods tracking	Data Collection

Improvement	Category
Disliked foods tracking	Data Collection
Dietary restrictions	Data Collection
Equipment availability tracking	Data Collection
Occupation tracking	Data Collection
Progressive onboarding wizard	UX Design
Health summary before dashboard	UX Design

9. Complete System Architecture

This architecture is unchanged from the original document. Phase 1 only touches the Frontend layer.

React Frontend
FastAPI Backend
MongoDB Database
Gemini / OpenAI API

10. Backend Modules (Planned — Phases 3-6)

Module	Endpoints	Purpose	Phase
1 — Auth Service	POST /signup POST /login POST /logout	User authentication with JWT	Phase 5
2 — User Service	GET /user PUT /user DELETE /user	Manage user profile	Phase 3
3 — Health Profile	POST /health-profile GET /health-profile	Store health metrics	Phase 3
4 — AI Recommendation	POST /generate-meal-plan POST /generate-workout	Talks to Gemini/OpenAI API	Phase 6

Module	Endpoints	Purpose	Phase
5 — Progress Tracker	POST /weight-entry GET /weight-history	Daily weight logs	Phase 7

11. Database Design (Updated Schema)

The Profiles collection has been updated to include all new fields collected during Phase 1. Other collections are unchanged.

Users Collection

```
{
  "_id": "",
  "name": "",
  "email": "",
  "password": ""
}
```

Profiles Collection (Updated — includes Phase 1 fields)

```
{
  "_id": "",
  "userId": "",
  "age": 20,
  "gender": "",
  "height": 157,
  "weight": 100,
  "goalWeight": 75,
  "foodPreference": "",
  "favouriteFoods": [],
  "dislikedFoods": [],
  "dietaryRestrictions": [],
  "monthlyFoodBudget": 0,
  "activityLevel": "",
  "availableEquipment": [],
  "occupation": "",
  "lifestyle": ""
}
```

12. Phase Breakdown

Phase	Focus	What Was / Will Be Built	Output
1 - Done	Foundation (Week 1)	Landing, auth pages with validation, 4-step onboarding wizard, health calculations, localStorage persistence, live dashboard cards	Functional Frontend UI

Phase	Focus	What Was / Will Be Built	Output
2	React (Week 2)	Rebuild Phase 1 in React: components, routing (React Router), state management	Frontend prototype
3	FastAPI (Week 3)	Backend server, API endpoints, CRUD operations, JSON handling	Working API
4	Database (Week 4)	MongoDB: users, profiles, conditions, logs	Persistent data
5	Auth (Week 5)	JWT authentication, password hashing, protected routes	Secure login
6	AI Layer (Week 6)	Gemini/OpenAI: meal plan generation, workout generation, prompt engineering	AI recommendations
7	Progress (Week 7)	Weight logs, progress dashboard, charts (streaks, water, weight trend)	User analytics
8	Deploy (Week 8)	Vercel (frontend), Render (backend), MongoDB Atlas (database)	Live app

13. Future Versions (Version 2)

After the MVP is live and validated, the following features are planned:

- Multiple diseases — Diabetes, Thyroid, Hypertension
- Menstrual cycle tracking
- Grocery list generation
- AI chatbot
- Habit tracker
- Sleep tracking
- Notification system
- Doctor dashboard
- Community section
- Voice assistant